Speed−up per step against Scanpy with Dask, full data set

In−memory Scanpy dies at load here; Dask is the Scanpy path that finishes. Only runs that completed.

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	umap	tsne_default	tsne_fft	total (no tSNE)
bixverse mtx	0.4x	2.0x		5.1x	3.9x	1.3x	1.4x	3.7x	6.9x	5.3x	2.1x	2.4x
bixverse h5ad	0.5x	1.7x		5.0x	4.0x	1.2x	1.4x	3.6x	6.9x	5.2x	2.2x	2.8x
bixverse GPU*	0.6x	3.0x		5.5x	5.0x	2.1x	2.7x	4.4x	63x	6.1x	12x	4.8x
bixverse MLX*	0.9x	3.2x		6.0x	7.4x	4.0x	3.0x	4.4x	67x	6.3x	13x	6.0x
Seurat BPCells	0.3x	0.8x	lazy	0.3x	0.02x	0.02x	0.5x	0.1x	0.8x	1.3x	1.1x	0.2x

bixverse normalises during load; Dask defers normalisation, so the normalise column carries no fold change.
Dask has no DGE step (Wilcoxon fails on Dask arrays), so its total lacks one; in−memory Scanpy's takes 602 s at 500k.
bixverse CPU tSNE builds its kNN with IVF.